

Luba Ira Korlat

Software Engineer | Python | TypeScript | Flutter | Backend | AI/NLP

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Professional Summary

Software Engineer and registered freelance developer with an M.Sc. in Software Engineering and hands-on delivery across Python, TypeScript, Dart/Flutter, backend APIs, mobile apps, and web projects. Build practical software from requirements to tested releases, with experience in AI/NLP, numerical modeling, databases, debugging, and technical teaching. Known for quickly understanding complex systems, explaining technical ideas clearly, and turning ambiguous problems into working implementations.

Technical Skills

Languages: Python, TypeScript, Dart, Java, C, C++, C#, SQL, R, HTML, CSS

Backend and APIs: FastAPI, Django, Node.js, NestJS, REST APIs, authentication concepts, testing

Frontend and Mobile: Flutter, React, React Native, Next.js, Phaser, Android development

AI, NLP, and Data: TensorFlow, Keras, PINNs, NLP, RAG concepts, OpenCV, model evaluation

Databases and Tools: PostgreSQL, MySQL, MongoDB, Firebase, Supabase, CouchDB, Git, Docker, Linux, CI/CD

Work Experience

Software Developer | Self-Employed | Beersheba, Israel | January 2025 - Present

- Deliver mobile apps, websites, backend systems, and technical prototypes for private clients as a registered Sole Proprietor.
- Build full-stack and Flutter solutions using Python, Dart, databases, APIs, and deployment-oriented project workflows.

Instructor | Cyber Education Centre | Beersheba, Israel | November 2025 - Present

- Teach networks, sockets, client-server programming, and cybersecurity fundamentals in the Magshimim Program.
- Review student code and provide direct feedback on correctness, debugging, protocol logic, and software structure.

Computer Science and Mathematics Tutor | Alut Organization and Ono Academic College | Beersheba, Israel | January 2026 - Present

- Teach computer science and mathematics to students on the autism spectrum using structured explanations and practical examples.

Tutor and Academic Transcriber | Aleh Association | Beersheba, Israel | September 2023 - December 2025

- Tutored students with disabilities in software engineering, programming, mathematics, and academic technical subjects.
- Created accessible summaries and clarified course material so students could follow complex technical content independently.

Software Engineering Tutor | Sami Shamoon College of Engineering | Beersheba, Israel | April 2023 - May 2024

- Taught algorithms, data structures, object-oriented programming, and core software engineering topics to undergraduate students.

Database Practice Tutor | Achva Academic College | Beersheba, Israel | November 2021 - April 2022

- Led SQL, relational modeling, and normalization practice sessions for Information Systems students.

Selected Projects

HeatEquationSimulator | Python, TensorFlow, Keras, CUDA, Tkinter, PINNs

GitHub: github.com/lubacareer/HeatEquationSimulator

- Developed a desktop simulation tool for the heat equation and two-phase Stefan problem with PINNs, numerical methods, analytical baselines, and a Tkinter interface.

ExitViolet | TypeScript, Phaser, Vite, Playwright

GitHub: github.com/lubacareer/ExitViolet

- Built a browser-first isometric psychological thriller vertical slice with menu overlays, opening video flow, room exploration, blockers, interactive hotspots, asset processing, and Playwright smoke tests.

The Case of the Missing Logic (CML) | TypeScript, Phaser, Vite

GitHub: github.com/lubacareer/CML

- Implemented a 2D point-and-click comedy mystery slice with scene navigation, character movement, branching dialogue, inventory, map gating, item use, and a playable puzzle chain.

spritebury | TypeScript, Next.js, Supabase

GitHub: github.com/lubacareer/spritebury

- Prototyped an early-stage browser virtual city and social web game concept focused on accounts, pixel avatars, profiles, inventory ownership, shops, jobs, homes, presence, and server-authoritative actions.

ast2 | Dart, Flutter, Firebase, AI integrations

GitHub: github.com/lubacareer/ast2

- Built an all-in-one astrology, Feng Shui, and crystal identification app with AI-generated interpretations, local fallback content, Firebase-backed flows, and a local vision-service integration.

ccomp | C, Flex, Bison, compiler construction

GitHub: github.com/lubacareer/ccomp

- Implemented a teaching-oriented C compiler that lexes and parses a C subset, builds an AST, performs semantic checks, emits TAC, and generates x86-64 AT&T assembly.

IRCourse | Python, machine learning, NLP

GitHub: github.com/lubani/IRCourse

- Completed academic information-retrieval and NLP work including sentiment analysis with Logistic Regression and Naive Bayes, plus an abstractive summarization flow using BART.

Education

Master of Science in Software Engineering | Sami Shamoon College of Engineering | Beersheba, Israel

Software Engineering, graduated cum laude | GPA: 96/100 | 2022 - 2024

Thesis: 2-Phase Stefan Problem for Lunar Regolith Melting. Relevant coursework: Deep Learning, Advanced Algorithms, Natural Language Processing, Computer Vision, Advanced Databases.

Bachelor of Science in Software Engineering | Sami Shamoon College of Engineering | Beersheba, Israel

Software Engineering | GPA: 80/100 | 2018 - 2022

Final project: Hebrew text summarization using natural language processing.

Research Experience

Research Student | Sami Shamoon College of Engineering | Beersheba, Israel | 2022 - 2024

- Developed Python simulations, numerical methods, and Physics-Informed Neural Networks for heat-transfer and phase-change research.

Military Service

National Service Secretary | Beersheba Magistrate Court | Beersheba, Israel | September 2011 - September 2012

- Completed one year of Sherut Leumi National Service in an administrative role using court information systems and legal-administrative workflows.

Languages

English: Fluent | Hebrew: Fluent | Russian: Fluent